

Abstract

How much embedding dimensionality is needed for a given corpus? Can a model trained on a simpler domain transfer effectively to semantically richer tasks?

To answer these questions, we study how domain and embedding dimensionality interact in unsupervised sentence encoders. Using two extreme corpora, **Taylor Swift lyrics** and **Nakul Verma's papers**, we train **contrastive BERT models from 4D to 767D** and evaluate geometry and transfer learning properties.

Surprisingly, lower-dimensional models trained on lyrics (4–256D) outperform paper-trained models on STS-B, with papers only catching up at 767D. This suggests that simple, high-frequency text can be an efficient pretraining source for compact encoders, while complex text needs far more dimensions to expose useful structure.

Methodology

Base model: BERT-base-uncased with mean pooling. A projection head maps pooled representations to a target dimension $d \in \{4, 16, 64, 256, 767\}$ followed by L_2 normalization.

Training domains: Each song or each paper section is treated as one document, and neighboring sentences within a document form positive pairs. The corpora are token-matched for training compute parity:

- 1) Taylor Swift lyrics (short, repetitive, low-entropy)
- 2) Nakul Verma's ML papers (long, information-dense)

Training objective: We train unsupervised SimCSE-style contrastive models with multiple positive examples, sampling local neighbors within each document. Following Wang et al. (2023), we use a two-step training scheme:

Step 1. Freeze pooler, train encoder (3 epochs).

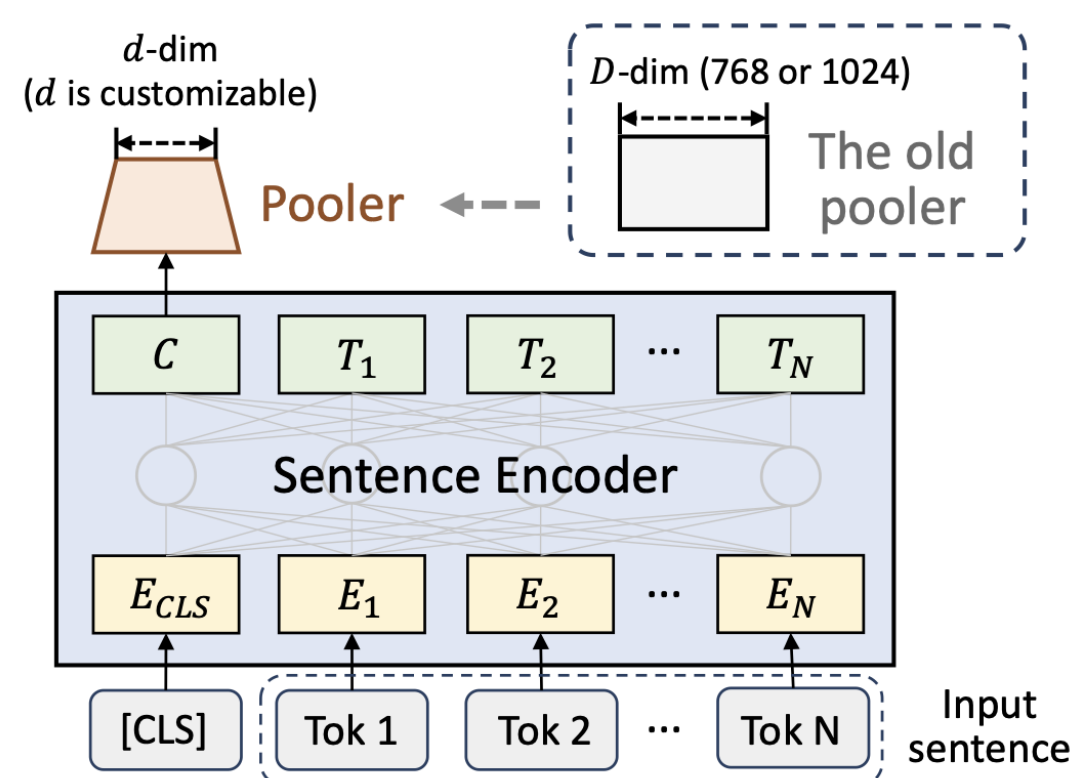
Step 2. Freeze encoder, train pooler (3 epochs).

PCA projections of the frozen BERT encoder at the same target dimensions are used as a baseline.

Evaluations

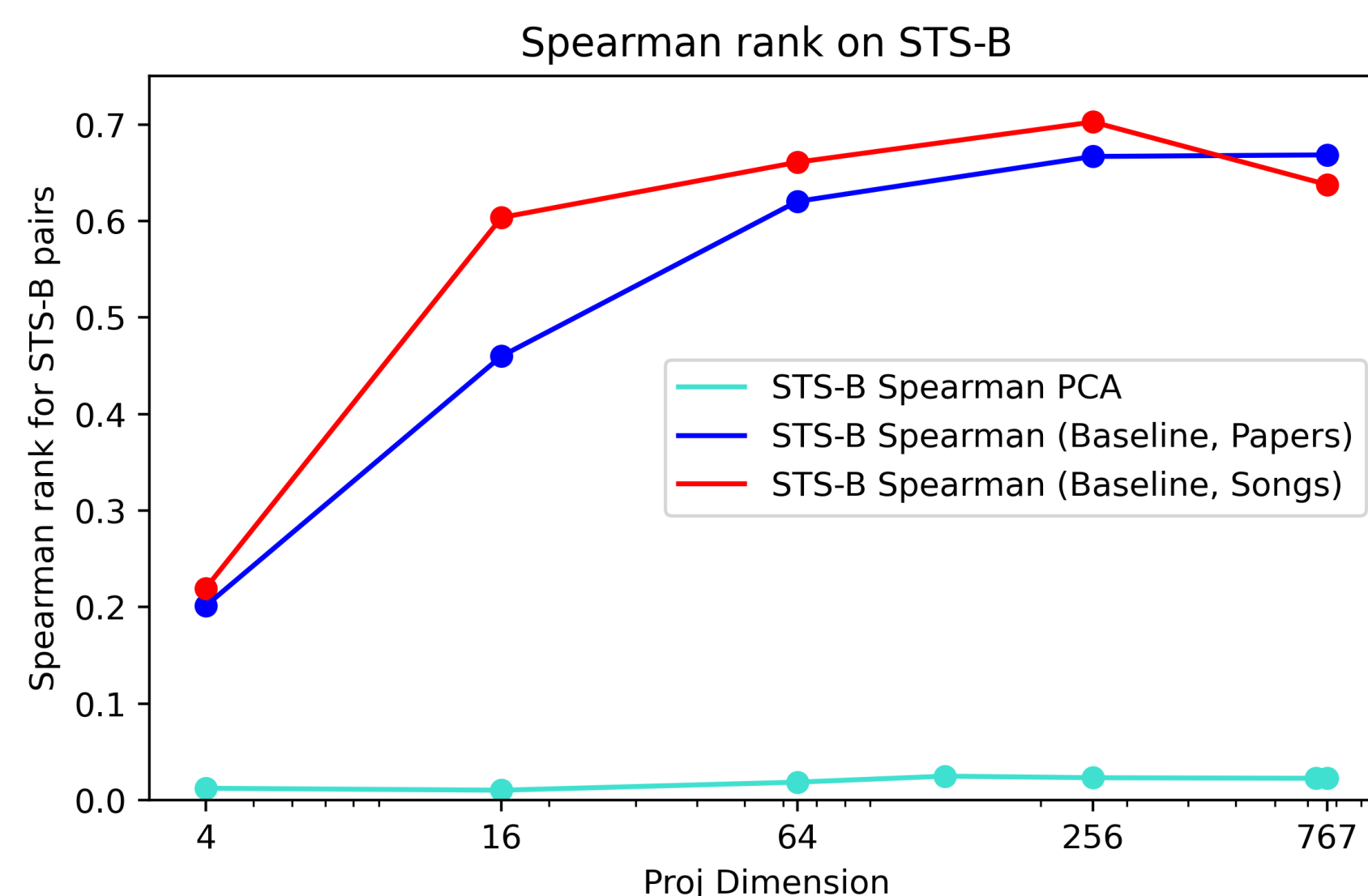
Geometry: 1) Alignment and uniformity losses (Wang et al., 2020) on in-domain and out-of-domain test sets. Alignment measures how close positive pairs are, and uniformity captures how evenly embeddings are spread on the hypersphere. 2) t-SNE embeddings of sentences from songs and papers.

Transfer learning: zero-shot STS-B Spearman.



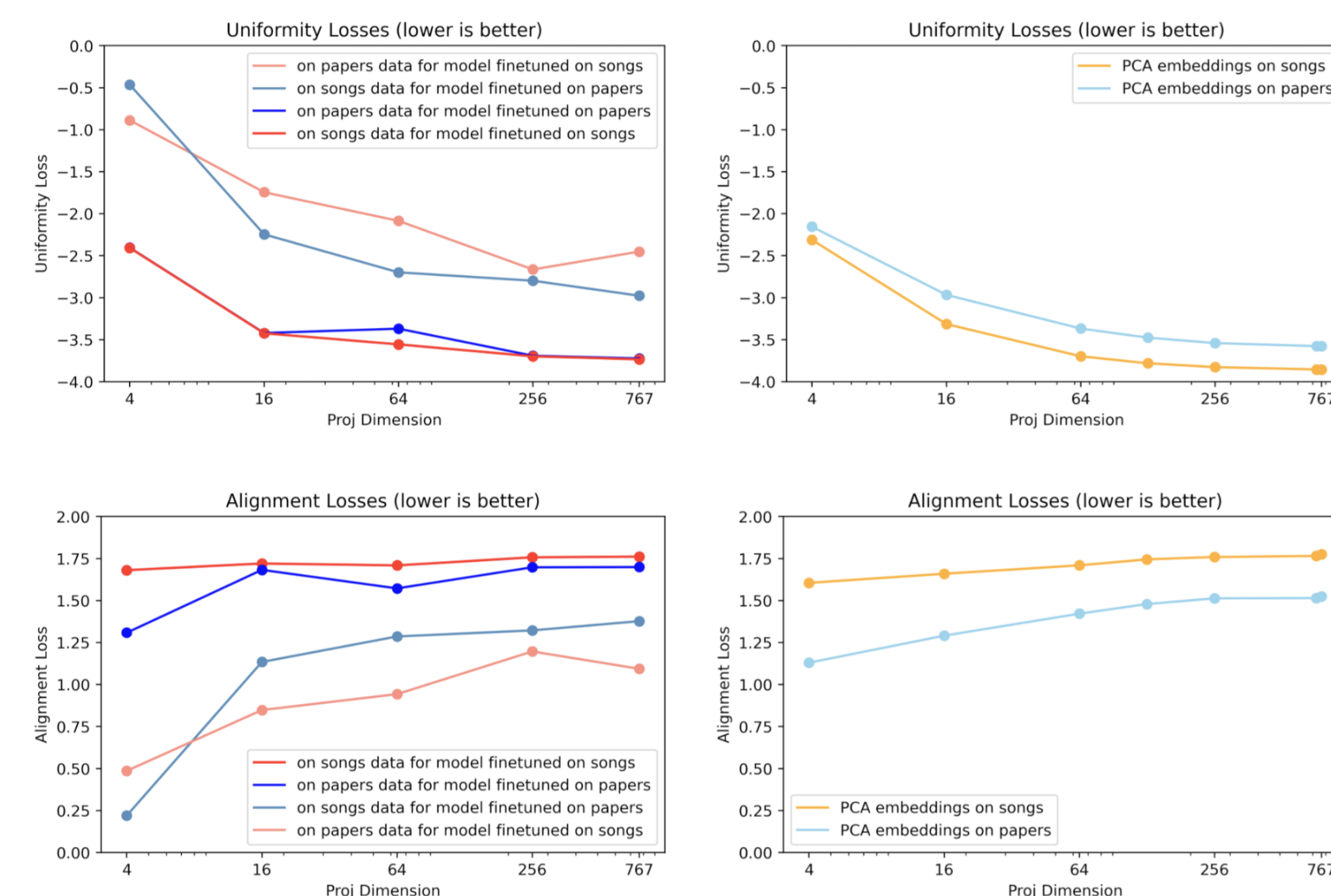
Results

Spearman Rank on STS-B: Lyrics-trained encoders rise sharply between 4D and 256D, peaking long before the full BERT size of 768D. Paper-trained models, in contrast, only come to full potential at 767D, hinting at a much **higher intrinsic dimensionality of scientific text**. The PCA baseline stays near zero Spearman rank across all projection sizes, suggesting that **linear compression alone cannot expose semantic structure** – the pooler must learn it.

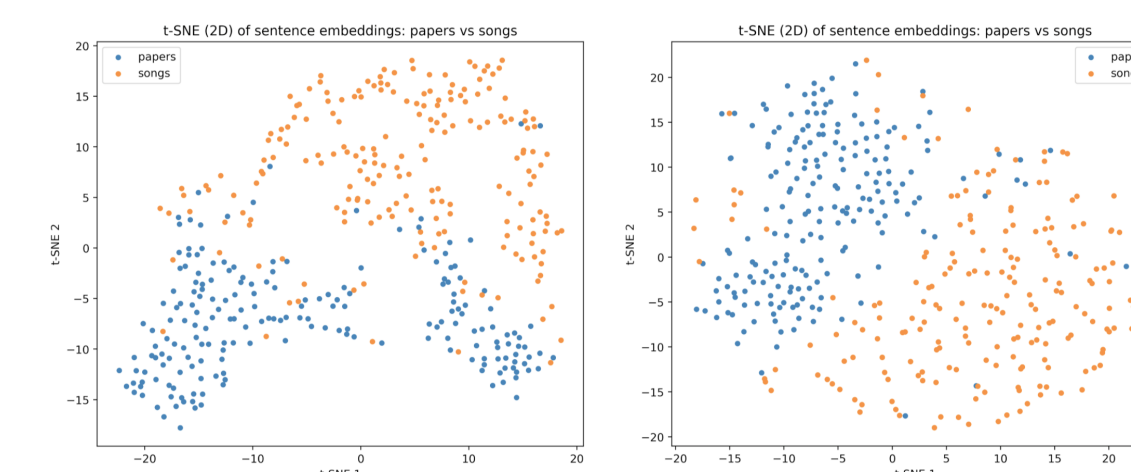


Uniformity and Alignment: All plots show the **alignment vs uniformity tradeoff**: higher dimensions improve uniformity but degrade alignment. For PCA specifically, uniformity losses are much smaller than alignment losses for all dimensions, indicating that embedded sentences occupy the full hypersphere but do not preserve local similarity structure.

For contrastive training, **in-domain data** (eg. songs embedded with a model trained on papers) **becomes well-spread** (good uniformity) but not overly tight, and **out-of-domain data** often collapses, achieving **artificially low alignment with poor global geometry**. In addition, papers-trained models behave more similarly in terms of both losses on different domains, while songs-trained models have larger oscillations when domains are switched.

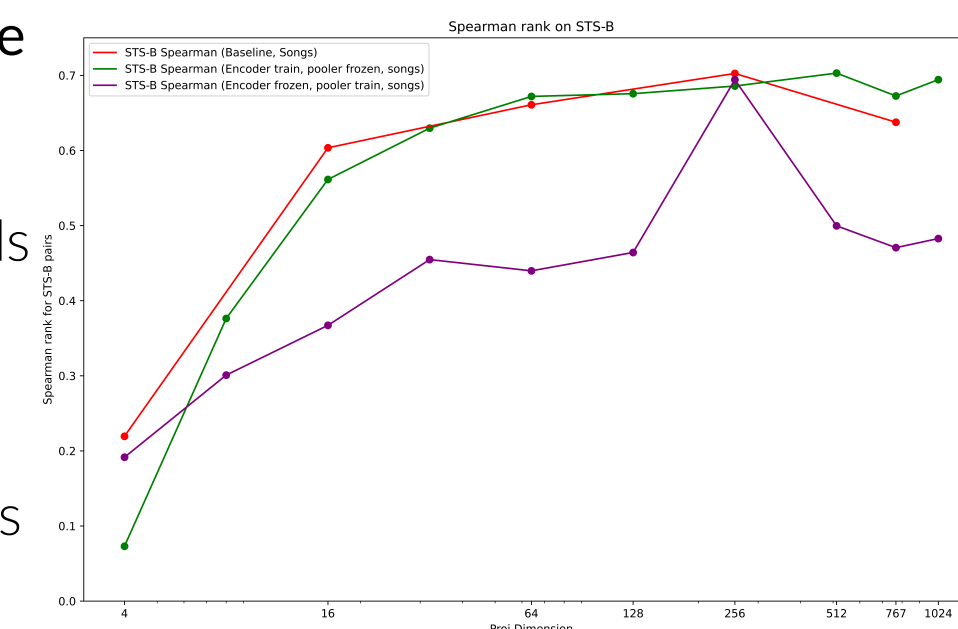


Songs and Papers Clusters Embedded with t-SNE: Even with only 4 dimensions, the embeddings already separate lyrics from papers as domain identity is cheap to encode. But it is only at 16+ dimensions that we see rounder clusters, more stable distances, and less distorted structure. This **mirrors the rise in STS-B performance**: the transition from 4D to 16D is where real semantic structure begins to emerge. The plots below are for our BERT+pooler model trained on songs.



Ablation on Training Procedure

on Songs-Trained Model: For smaller dimensions (4, 16), the two-step training process yields the best Spearman rank correlations. However, at certain higher dimensions training only the encoder yields the best results. Training only the pooler is not sufficient for learning useful representations.



Takeaways and Future Work

- **Lyrics-trained models dominate at low dimensions (4–256D) on STS-B**, while papers-trained models only overtake at 768D ⇒ **scientific text appears inherently higher-dimensional**.
- **Uniformity and alignment conflict** with each other, with **papers-trained models being more stable** when transitioning from in-distribution to out-of-distribution data.
- **Low-dimensional embeddings** are able to **separate domains (even at 4D)**, but more meaningful local structure emerges only at 16D or more.
- Linear compression (**PCA**) is **insufficient for transfer learning** tasks in our experiments: contrastive learning is necessary to expose semantic structure even at small dimensions.

For future work, we would like to explore **additional domains** (such as news, code comments etc.), try other **embedding models beyond BERT**, and use **alternative training objectives** (for instance, non-contrastive methods).

Main References

- [1] Wang, Hongwei, Hongming Zhang, and Dong Yu. "On the dimensionality of sentence embeddings." arXiv preprint arXiv:2310.15285 (2023).
- [2] Wang, Tongzhou, and Phillip Isola. "Understanding contrastive representation learning through alignment and uniformity on the hypersphere." International conference on machine learning. PMLR, 2020.
- [3] Reimers, Nils, and Iryna Gurevych. "Sentence-bert: Sentence embeddings using siamese bert-networks." arXiv preprint arXiv:1908.10084 (2019).
- [4] Gao, Tianyu, Xingcheng Yao, and Danqi Chen. "Simcse: Simple contrastive learning of sentence embeddings." arXiv preprint arXiv:2104.08821 (2021).

